

MEASUREMENT OF CARDIOTHORACIC RATIO AND DETECTION OF CARDIOMEGLALY IN X-RAY IMAGES USING DEEP LEARNING

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Abstract:

In this study, the cardiothoracic ratio is automatically measured by extracting lung and heart regions in a chest X-ray image and measuring their widths. The proposed method uses a deep learning model based on U-Net++ with VGG19_bn encoders. The results of cardiothoracic enlargement detection using the cardiothoracic ratio measured by the proposed method showed a high degree of agreement with the judgment made by a physician. As a result, the automatic cardiothoracic ratio measurement system using the proposed method contributes to a significant reduction in the time and labor of physicians.

Keywords:

Chest X-ray images; Cardiothoracic ratio; Cardiomegaly; Deep learning; segmentation

1. Introduction

The number of patients with heart failure is increasing in Japan, a hyper-aged society. On the other hand, as an effective preventive method for heart failure, physicians measure the width of the heart and rib cage in chest X-ray images taken during chest X-ray examinations and determine the presence of cardiac enlargement using the cardiothoracic ratio (CTR) calculated from these widths [1]. However, because physicians must manually label the lung and heart regions, errors are likely to occur, and incorrect interpretations may be made [2]. Therefore, an automated CTR measurement technique that is faster and more accurate is required.

1.1. Cardiothoracic Ratio

Chest X-ray images not only allow subjective assessment of heart size, but also allow measurement of the heart's contour, which may be related to the size of the thorax. Studies have also shown that there is a sufficient correlation (>0.7) between the size of the heart measured on chest X-ray images and that measured at autopsy [2]. Therefore, chest X-ray images are the most commonly used to detect cardiac

enlargement, and the cardiothoracic ratio (CTR), which evaluates the size of the heart and rib cage, is also an important index for determining cardiac enlargement. The cardiothoracic ratio is the ratio of the width of the heart to the width of the thorax. The cardiothoracic ratio can be calculated using Equation (1).

$$CTR = \frac{MLD + MRD}{ID} \quad (1)$$

As shown in Figure 1, ID (Inner Diameter) is the largest horizontal width of chest. MLD (Maximum Left Diameter) and MRD (Maximum Right Diameter) are the distance from spine midline to most lateral left ventricle and right ventricle, respectively [2,3]. The definitions of the midline spine and the left and right ventricle are shown in Fig. 2.

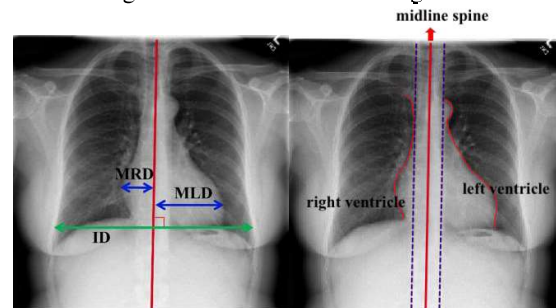


FIGURE 1. CTR

FIGURE 2. Definitions

1.2. Automated CTR Calculation

Currently, there are many systems and algorithms that can automatically calculate CTR. Konica Minolta, Inc. has developed a system that can calculate the thoracic and cardiac widths necessary for cardiothoracic ratio measurement by applying its thoracic lung field extraction technology and display the cardiothoracic measurement line following the slope of the lung field and the calculated measurement values. In a recent study, a high accuracy of 87

[%] was obtained in the detection of cardiac enlargement in 600 images selected from a chest X-ray image database called ChestX-ray8 published by the NIH using the transfer deep learning method [4]. Therefore, this study focuses on image segmentation methods based on deep learning, and furthermore, an automatic CTR measurement method using cardiothoracic measurement lines following the slope of the segmented lung fields.

2. Method

2.1. Dataset Preparation

In order to segment the heart and lung regions in chest X-ray images using deep learning, we first need to construct training datasets for the lung and heart segmentation models, respectively. To build the training dataset, we used the NIH ChestX-ray14 dataset published by the National Institutes of Health (NIH), which contains 100,000 anonymized chest X-ray images and labels of disease names determined by radiologists. We used a dataset of chest X-ray images [5, 6].

In the NIH ChestX-ray14 dataset, we created a new chest X-ray image dataset by classifying images into those with a cardiac enlargement label and those with a cardiac enlargement negative label, respectively. Images with poor image quality and images with ambiguous boundaries between the heart and lungs were selected and removed. A total of 367 chest X-ray images were then obtained, including 201 chest X-ray images labeled cardiac enlargement and 166 chest X-ray images labeled cardiac enlargement negative, for use in the heart segmentation model. A total of 407 chest X-ray images were acquired, including 197 chest X-ray images with cardiac enlargement labels and 210 chest X-ray images with cardiac enlargement-negative labels for the lung segmentation model.

Next, to create a training dataset for the segmentation model, we annotated the heart and lungs in the selected chest X-ray images using the annotation tool LabelMe, which is available in python environments such as Anaconda [7]. As a result, a total of 367 heart mask images and 407 lung mask images were obtained. Then, 294 and 73 images were randomly segmented from the cardiac mask images and 326 and 81 from the lung masks, respectively, as the training and validation datasets for training and validation, respectively.

In order to verify the accuracy of the CTR calculated from the segmented lung and heart regions, it is necessary to prepare a validation dataset in addition to the training dataset. Therefore, 296 images with cardiac enlargement label and 301 images with cardiac enlargement negative label were randomly obtained from the new dataset described above. As image preprocessing, we standardized the image size to 256

[pixel]×256 [pixel] and normalized the images to the prepared dataset.

2.2. Segmentation Models

U-Net is a model for semantic segmentation developed by Olaf et al. for biomedical applications [8]. FPN is a pyramid structure model often used for multi-scale object detection [10]. FPN is also used for semantic segmentation because of its similarity to the U-Net network structure [9]. In this study, we compare and evaluate three approaches for heart and lung segmentation: U-Net, U-Net++ which is based on U-Net and FPN [8]. In this study, the VGG19_bn architecture, which has more convolutional layers, is used as the encoder instead of VGG11. For ease of comparison, the encoders for the other two models are also based on the VGG19_bn architecture.

2.3. Valuation Index

In this study, it is necessary to measure the width of the extracted lung and heart contours. Therefore, we evaluate the accuracy by using the IoU, which can evaluate the contours rigorously. The formula for calculating IoU is shown in Equation (2). The definitions of TP, FP, and FN, which constitute equation (2), are shown in Table 1.

$$IoU = \frac{TP}{TP + FP + FN} \quad (2)$$

TABLE 1. Definition of a confusion matrix

		True class	
		Positive	Negative
Predicted class	Positive	TP(True Positive)	FN(False Negative)
	Negative	FP(False Positive)	TN(True Negative)

2.4. Postprocessing

Because the lung and heart masks extracted by segmentation contain holes and noise, image post-processing is required to remove the noise. In this research, image post-processing is performed using morphological operations. In the mask image of the lungs, the two largest connected components are used as the lung region, and the other components are removed as noise. In the case of the heart, the image post-processing is performed by setting the connected component at the center of the image to the heart region. Figs. 3 and 4 show the mask lung images before and

after postprocessing. Figs. 5 and 6 show the heart mask images before and after postprocessing.

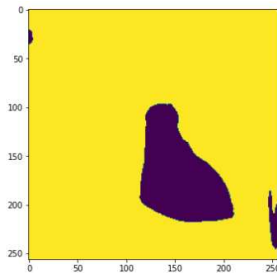


FIGURE 3. postprocessing

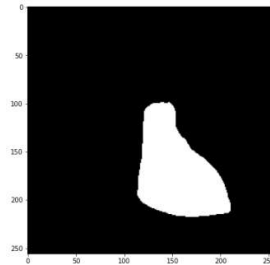


FIGURE 4. After postprocessing

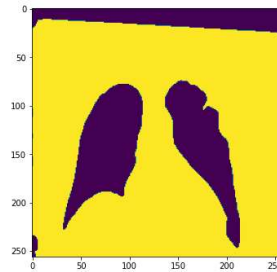


FIGURE 5. postprocessing

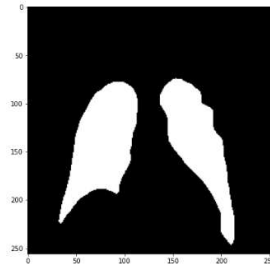


FIGURE 6. After postprocessing

2.5. Cardiothoracic Ratio Calculation

To calculate the cardiothoracic ratio, it is necessary to measure the width of the lung and heart regions. The width of the heart is calculated by summing the maximum distance from the spine midline to the left and right cardiac edges, but depending on the patient's imaging posture, the lung field and heart may be tilted due to the patient's body tilt, which may interfere with the measurement. Therefore, to measure the width of the heart, the spine midline must be displayed as the CTR measurement line in the mask image. Therefore, we propose a method in which the CTR measurement line is a straight line connecting the intersection of the two smallest circumscribing circles of the two connected components in the mask image of the lungs. Fig. 10 shows the CTR measurement line displayed on the mask image using this method. The minimum and maximum points of the mask in the vertical axis of the measurement line are found, and the sum of the distances from the minimum and maximum points to the measurement line is calculated. The sum of the distances calculated for the mask image of the heart is the width of the heart, and the sum of the distances calculated for the mask image of the lungs is the width of the rib cage. The measurements of the width of the thorax and heart using the CTR measurement lines are shown in Figs. 8-9.

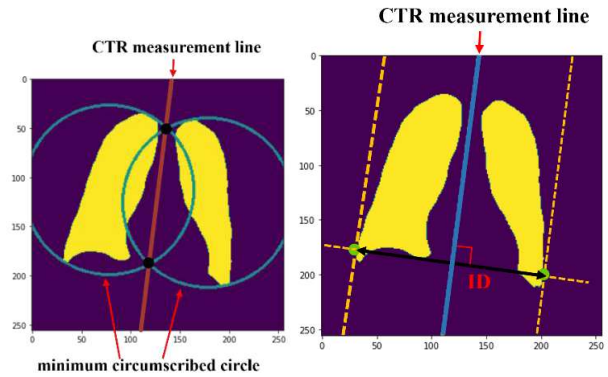


FIGURE 7. Minimum circumscribed circle

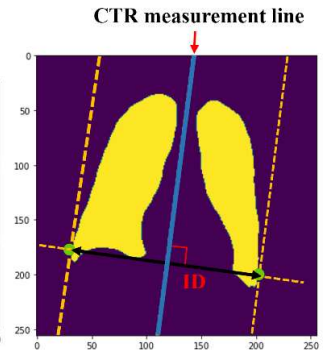


FIGURE 8. Lung mask

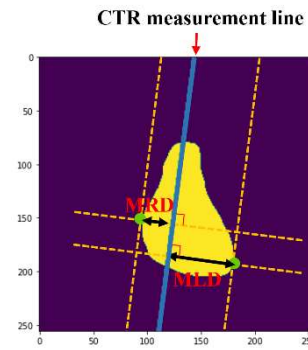


FIGURE 9. Heart mask

2.6. Cardiomegaly Detection

Cardiac hypertrophy is determined by the CTR value, and a cardiothoracic ratio of 0.5 or greater is generally considered to indicate cardiac hypertrophy. However, this value is not always correct, and is influenced by cardiac and non-cardiac factors such as examination technique, chest changes and heart rate. Therefore, it is necessary to find a reference value for judging cardiac hypertrophy that is compatible with the method proposed in this study [4]. Therefore, in this study, the ROC curve, which is commonly used in medical research, is used to determine the threshold value of cardiac enlargement CTR. The vertical axis of the ROC curve is the sensitivity (TPR: True Positive Rate) at a certain threshold value and the horizontal axis is the specificity (FPR: False Positive Rate), and the formulas for calculating sensitivity and specificity can be expressed by Equations (3-4). In this study, the definitions of TP, FN, FP and TN in the formulas are shown in Table 2.

$$TPR = \frac{TP}{TP + FN} \quad (3)$$

$$FPR = \frac{FP}{FP + TN} \quad (4)$$

TABLE 2. Definition of TP, FN, FP and TN

		Judgement result by threshold	
		cardiomegaly	no cardiomegaly
Expert label	cardiomegaly	TP(True Positive)	FN(False Negative)
	no cardiomegaly	FP(False Positive)	TN(True Negative)

3. Experiments

3.1. Segmentation Comparison

In this study, we compared three models in our segmentation method for chest X-ray images: a uniform training dataset, an FPN trained with hyperparameters and a decoder structure, U-Net and U-Net++, using the average IoU values on the validation dataset. Then, 296 chest X-ray images with cardiac enlargement label and 301 chest X-ray images with cardiac enlargement negative label were segmented using U-Net++. The heart and lung mask images obtained from the segmentation were post-processed using the technique described above. The post-processed heart and lung mask images were used as output images for subsequent cardiothoracic ratio measurements.

3.2. Cardiothoracic Ratio Calculation

The CTR measurement lines were detected in the mask image of the lungs using the minimum circumscribed circle method that was described above and were also displayed in the chest X-ray image and the mask image of the heart. Then, using the MLD, MRD and ID measured by the CTR measurement lines, the CTR was calculated according to Equation (1).

3.3. Cardiomegaly Detection

Images with a calculated CTR value of 0.5 or higher were judged to have cardiac hypertrophy. The accuracy and their average rate in images labeled with cardiac enlargement and those labeled negative for cardiac enlargement were then calculated according to the results of the determination. However, since the value of 0.5 is not always correct for cardiac and non-cardiac factors, an appropriate CTR threshold was obtained using ROC curves. The accuracy and their mean rates in the cardiac hypertrophy labeled and cardiac hypertrophy negative labeled images were again

calculated according to the results of cardiac enlargement determined using the appropriate CTR threshold.

4. Results

4.1. Segmentation Comparison

Table 3 shows the results of comparing the mean IoU values of the three segmentation methods, FPN, U-Net, and U-Net++, on the lung and heart validation data sets. In addition, the IoU here is averaged from 40 to 60 in the lung and heart validation datasets, since all three models begin to converge at epochs after 40. Some of the segmentation results for the heart and lungs for each method are shown in Fig. 13.

TABLE 3. Comparative evaluation results for IoU

Segmentation model	Mean IoU on validation datasets	
	lung	heart
FPN	0.966	0.983
U-Net	0.965	0.982
U-Net++	0.968	0.984

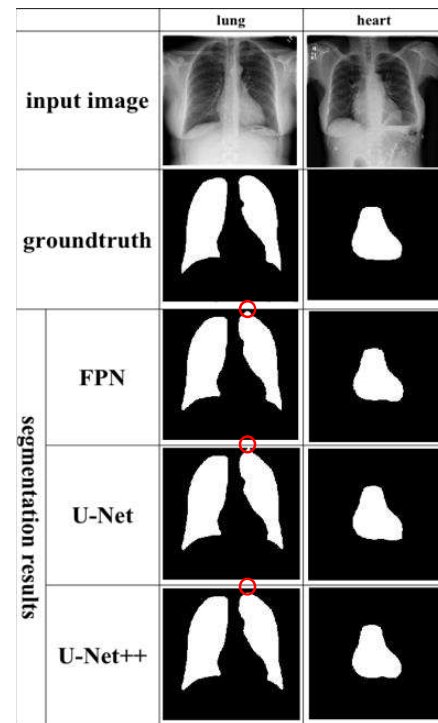


FIGURE 10. Segmentation results in heart and lungs for each model

4.2. Cardiothoracic Ratio Calculation

Some of the results of the ideally detectable CTR measurement lines in the cardiac enlargement labeled and cardiac enlargement negative labeled lung mask images are shown in Fig. 14 and Fig. 15 for the chest X-ray image and the heart mask image, respectively. Fig. 16 shows a part of the results when the CTR measurement line in the cardiac enlargement negative labeled image is shifted to the mid-thoracic vertebral line.

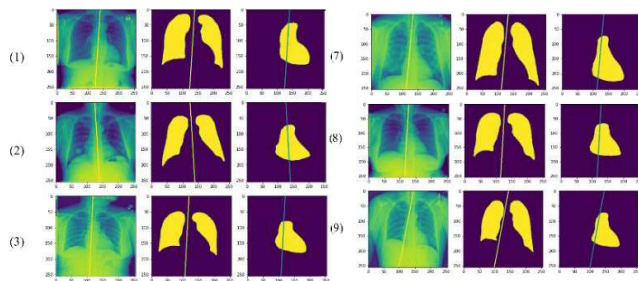


FIGURE 11. Positive labeled

FIGURE 12. Negative labeled

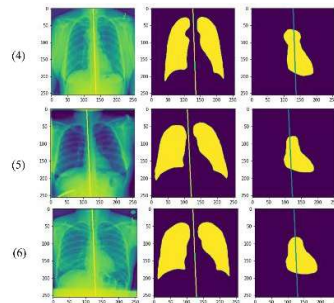


FIGURE 13. Negative labeled

4.3. Cardiomegaly Detection

Table 4 shows the results of cardiac enlargement when the CTR value is judged to be 0.5 or higher and the accuracy of the ROC curve is shown in Fig. 17. The results of cardiac enlargement detection using 0.49 as the threshold of CTR and its accuracy are shown in Table 5.

TABLE 4. Results and accuracy with a threshold of 0.5

		Judgement Results		Total [sheets]	Accuracy rate [%]	Mean Accuracy rate [%]
		Cardiomegaly [sheets]	No Cardiomegaly [sheets]			
label	cardiomegaly	249	47	296	83.78	85.75
	No cardiomegaly	37	264	301	87.71	

TABLE 5. Results and accuracy with a threshold of 0.49

		Judgement Results		Total [sheets]	Accuracy rate [%]	Mean Accuracy rate [%]
		Cardiomegaly [sheets]	No Cardiomegaly [sheets]			
label	cardiomegaly	270	26	296	91.22	87.14
	No cardiomegaly	51	250	301	83.06	

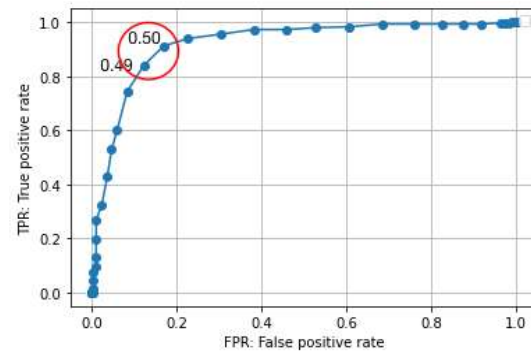


FIGURE 14. ROC curve

5. Discussion

5.1. Segmentation Comparison

In this experiment, Comparing the average IoU values when using U-Net++ and FPN, Table 1 shows that U-Net++ had higher IoU values. In addition, Table 1 shows that U-Net++ produced higher average IoU than FPN. The reason for this is thought to be that U-Net++ compensates for the inferiority of FPN by providing a deeper connection between the decoder and encoder of U-Net, and U-Net++ also demonstrates the superiority of U-Net for segmentation in medical images with small amounts of data. As shown in Fig. 13, U-Net++ is most reliable than FPN and U-Net for segmenting the boundaries between the lungs and the background in the lung segmentation results.

Therefore, in this experiment, U-Net++ is considered the best segmentation method for this study.

5.2. Cardiothoracic Ratio Calculation

In the measurement of the cardiothoracic ratio, as the chest X-ray images in Fig. 14 and Fig. 15 show, the CTR measurement lines detected in the mid-thoracic spine are well displayed by the method proposed in this study. In addition, the detected CTR measurement lines are accurately displayed according to the inclination of the lung fields in the images with scoliosis of the thoracic spine and in the chest

X-ray images where the lung fields and the heart are inclined due to the patient's posture.

As shown in Fig. 16, the detected CTR measurement line deviates from the midline of the thoracic spine due to the large difference in area caused by the asymmetry of the two lung areas.

5.3. Cardiomegaly Detection

The results in Table 4 and Table 5 indicate that a threshold value of 0.49 is more accurate than 0.5 in detecting cardiac enlargement. Furthermore, when using 0.49 as the threshold, the accuracy of detecting cardiac enlargement in images labeled as cardiac enlargement was 91.22 [%], which is a high ratio. Through this study, it can be concluded that there is a high degree of agreement between the judgment of cardiac enlargement by the CTR measured by the segmentation method and the judgment by the physician.

In addition, the judgment of cardiac enlargement on chest X-ray images of the same patient varied from physician to physician. Many physicians make judgments based on their own clinical experience, which may lead to individual differences. Therefore, this study, which quantitatively detects cardiac enlargement, has the potential to reduce the subjective diagnosis based on physicians' experience.

6. Conclusion

The objective of this study is to automatically measure the cardiothoracic ratio by extracting the lung and heart regions in chest X-ray images and measuring their widths using a deep learning segmentation method.

Based on the experimental results, U-Net++ is the best model for the segmentation method used in this study, so U-Net++ was used to segment the lungs and heart. The acquired mask images were used to measure the CTR, which is often used to detect cardiac enlargement. We then determined the threshold of CTR using ROC curves and detected cardiac enlargement. The results showed that the detection of cardiac enlargement using the method proposed in this study was highly consistent with the judgment of physicians.

For future work, it is first necessary to find a larger dataset of thoracic X-ray images to improve the accuracy and reliability of segmentation. In addition, the CTR measurement line detection method proposed in this study can still be improved, so it is necessary to reexamine the CTR measurement line detection method that can display the spine midline more correctly.

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